

RAG Chatbot - Detailed Improvement Analysis

CRITICAL ISSUES

1. Missing requirements.txt (currently empty).
2. Hardcoded Secret Key.
3. Chroma DB reset on every ingest causing data loss.
4. No error handling for uploads, DB, or LLM failures.
5. Database data loss when embeddings are recreated.

PERFORMANCE ISSUES

1. Blocking PDF processing.
2. No pagination for chat history.

CODE QUALITY ISSUES

1. Duplicate embedding initialization.
2. Missing input validation.
3. Inconsistent formatting and no type hints.
4. No structured logging.

FRONTEND ISSUES

1. No Enter key support.
2. Missing API error handling.
3. No loading states.
4. Upload success uses alerts.
5. No markdown rendering.

SUGGESTED IMPROVEMENTS

P0: requirements.txt, .env secrets, preserve Chroma DB, error handling.

P1: async processing, shared embedding utils, Enter key support, API error handling, pagination, metadata tracking.

WHAT'S WORKING WELL

- Clean frontend/backend separation.
- MMR retrieval.
- Simple database schema.
- Modern UI styling.
- OCR fallback for scanned PDFs.